

Trần Văn Huy 24it097

Bai 1.

```
huytran@DESKTOP-4VSRB7I:~$ ./cd ..
Ok, cwd is now: /home
Thuc hien boi Sinh vien: Tran Van Huy - MSSV: 24IT097
huytran@DESKTOP-4VSRB7I:~$
```

Bai 2.

```
huytran@DESKTOP-4VSRB7I:~$ gcc getwd.c -o getwd
huytran@DESKTOP-4VSRB7I:~$ ./getwd
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root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
landscape:x:101:103::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
huytran:x:1000:1000:/home/huytran:/bin/bash
TranVanHuy:x:1001:1002:/home/TranVanHuy:/bin/bash
huytran@DESKTOP-4VSRB7I:~$
```

Bai 3

```
huytran@DESKTOP-4VSRB7I:~$ nano groups.c
huytran@DESKTOP-4VSRB7I:~$ gcc groups.c -o gr
huytran@DESKTOP-4VSRB7I:~$ ./gr
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huytran adm cdrom sudo dip plugdev users docker
huytran@DESKTOP-4VSRB7I:~$
```

Bai 4.

```

huytran@DESKTOP-4VSRB7I:~$ nano rename.c
huytran@DESKTOP-4VSRB7I:~$ gcc rename.c -o rename
huytran@DESKTOP-4VSRB7I:~$ ./rename
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Usage: ./rename from to
huytran@DESKTOP-4VSRB7I:~$

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Bai 5.

```

huytran@DESKTOP-4VSRB7I:~$ nano sizeof.c
huytran@DESKTOP-4VSRB7I:~$ gcc- sizeof.c -o side
Command 'gcc-' not found, did you mean:
  command 'gcc' from deb gcc (4:15.2.0-5ubuntu1)
Try: sudo apt install <deb name>
huytran@DESKTOP-4VSRB7I:~$ gcc sizeof.c -o side
huytran@DESKTOP-4VSRB7I:~$ ./side
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int: 4
Fixed buffer, size known at compile time: "buf[8192]"
strlen: 0 - sizeof: 8192

Fixed-size, initialized buffer: "buf[2] = { 'a' }"
strlen: 1 - sizeof: 2

A pointer: 'char *s = "abcde"'
strlen: 5 - sizeof: 8

Another initialized buffer, implicitly sized: 'buf3[] = "abcde"'
strlen: 5 - sizeof: 6

A pointer to malloc(3)'d memory:
strlen: 0 - sizeof: 8

After strncat(3)ing "0123456789" to 's':
strlen: 10 - sizeof: 8

huytran@DESKTOP-4VSRB7I:~$

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Bai 6.

```

huytran@DESKTOP-4VSRB7I:~$ nano time.c
huytran@DESKTOP-4VSRB7I:~$ gcc time.c -o time
huytran@DESKTOP-4VSRB7I:~$ ./time
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time(3) says:      1789009419
gettimeofday(2) says: 1789009419.32686
clock_gettime(2) says: 1789009419.32692420

asctime(3) says:    Thu Sep 10 03:03:39 2026
localtime is:      Thu Sep 10 10:03:39 2026
time: unable to call mktime(): Success
huytran@DESKTOP-4VSRB7I:~$

```

Bai 7.

```

cat: wait_unlink.c: No such file or directory
huytran@DESKTOP-4VSRB7I:~$ nano wait_unlink.c
huytran@DESKTOP-4VSRB7I:~$ gcc wait_unlink.c -o wait
huytran@DESKTOP-4VSRB7I:~$ ./wait
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Available space is now:
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/sdd        1055762868 3078564 998980832   1% /

```